Illustration Notes

Notes on specific figures that need attention or have additional context.

Chapter 5 - Flow Diagrams with Unknown Origin

Several flow diagrams in Chapter 5 have unknown origins from "the original manuscript". These are currently marked as **Red** status.

Figures Needing Replacement

Figure	Description	Status
fig_05_04	Potential flow round a sheet	Red - Unknown origin
fig_05_05	Flow round a rotating cylinder	Red - Unknown origin
fig_05_06a	Airfoil with zero circulation (overview)	Red - Unknown origin
fig_05_06b	Trailing edge close-up (zero circulation)	Red - Unknown origin
fig_05_06c	Velocity around airfoil surface	Red - Unknown origin
fig_05_07a	Airfoil with circulation (overview)	Red - Unknown origin
fig_05_07b	Trailing edge close-up (with circulation)	Red - Unknown origin
fig_05_07c	Velocity around airfoil surface (circulation)	Red - Unknown origin
fig_05_08a	Flow around wing with flap (c=0)	Red - Unknown origin
fig_05_08b	Flow around wing with flap (c=6.1)	Red - Unknown origin
fig_05_09	Streamlines in ground effect	Red - Unknown origin
fig_05_12	Potential flow round a sail	Red - Unknown origin

Recently Replaced Figures (Pending Hugh's Review)

The following figures have been replaced with self-created Python versions. These need Hugh's review to confirm they are acceptable replacements.

Chapter 4

Figure	Description	Python Script	Notes
fig_04_09.png	Two-source wave interference pattern	`wave_interference.py`	Grayscale interference pattern from two coherent point sources

Figure	Description	Python Script	Notes
fig_04_12.png	3D Gaussian surface ("Mexican Hat")	`gaussian_3d_surface.py`	Rainbow-colored 3D surface with discrete color bands
fig_04_13.png	Rayleigh distribution (Probability vs Radius)	`fig_04_13_rayleigh.py`	$P(r) \propto r \times \exp(-2r^2/\sigma^2)$ curve
fig_04_14.png	Brewster angle / refraction diagram	`fig_04_14_brewster.py`	Shows incident, reflected, and refracted rays with polarization arrows

Chapter 7

Figure	Description	Python Script	Notes
fig_07_05.png	Lowest four oscillation modes for 2D membrane (drum skin)		Self-created replacement

All Python scripts are located in `/python_flow_figures/` directory.